Book Copy Circulation Guard

Assignment Problem 3

Concept and approach

Keep inventory fields private and protect the invariant 0 <= copiesAvailable <= copiesTotal. A checkout decrements only when at least one copy is available; a check-in increments only when some book is currently checked out. Invalid operations do nothing silently.

Important implementation notes

The assignment's second sample assumes three copies were checked out before calling checkIn four times; the main method explicitly demonstrates both the lower and upper boundaries.

Java solution

```
class BookInventory {
private final int copiesTotal;
private int copiesAvailable;

public BookInventory(int copiesTotal) {
if (copiesTotal <= 0 || copiesTotal > 500)
throw new IllegalArgumentException("copiesTotal must be between 1 and 500");
this.copiesTotal = copiesTotal;
this.copiesAvailable = copiesTotal;
}

public void checkOut() {
if (copiesAvailable > 0) copiesAvailable--;
}

public void checkIn() {
if (copiesAvailable < copiesTotal) copiesAvailable++;
}

public int getCopiesAvailable() {
return copiesAvailable;
}

public static void main(String[] args) {
BookInventory b = new BookInventory(3);
b.checkOut(); b.checkOut(); b.checkOut(); b.checkOut();
System.out.println(b.getCopiesAvailable());
b.checkIn(); b.checkIn(); b.checkIn(); b.checkIn();
System.out.println(b.getCopiesAvailable());
}
}
```

Sample output

```
0
3
```